

Exact Robust Local Graph-Smoother Design Under Hidden-Topology Completions

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Abstract—Graph Tikhonov smoothing is a global rational graph filter, yet a distributed processor may observe and communicate only on a labeled port set while the remaining topology and inputs are hidden. We design one common port-local linear operator with a uniform error certificate over all such completions. Finite completions may contain at most h hidden vertices, cycles, and nonnegative weights; the unbounded model permits arbitrarily many hidden vertices. We give an exact closed-convex-hull description of every port block of $(I+L)^{-1}$ using set partitions and integer hidden-budget allocations. Every retained vertex is exposed through connected maximum-degree-two path limits. Removing the hidden budget gives a partial-partition polytope, with exact operator-norm Hausdorff distance $q/(q+h)$ from its finite- h counterpart. Retaining hidden-input columns reduces finite-budget worst cases for all unitarily invariant norms to canonical scenarios. For Schatten exponents at least two, exact vertex pruning and an unbounded partial-partition reduction follow; a two-vertex counterexample proves that this exponent threshold is sharp. Spectral-norm design under any semidefinite-representable local rule is therefore a finite necessary-and-sufficient semidefinite program. On a fixed 54-problem theorem-matched grid, one-hop constant-preserving rules reduce the certified radius over zero-hop rules by a median 18.7%. In a representative four-port, one-hidden-node problem, the finite budget improves the radius over the unbounded model by 19.1%, while a positive-contraction relaxation incurs 10.2% exact-risk overhead. For the same problem, 30 same-distribution trials with 32 training completions show that random sampling underreports the exact radius by a median 18.0%. We use, but do not claim, the established random-forest mean-projection identity.

Index Terms—Distributed graph filtering, graph Tikhonov regularization, hidden nodes, robust optimization, semidefinite programming, random spanning forests.

I. INTRODUCTION

For a graph signal y , Tikhonov smoothing applies the resolvent of the graph Laplacian,

$$\begin{aligned} \hat{x}_G(y) &= \arg \min_x \left\{ \frac{1}{2} \|x - y\|_2^2 + \frac{1}{2} x^T L_G x \right\} \\ &= (I + L_G)^{-1} y. \end{aligned} \quad (1)$$

This stable low-pass graph filter is global: changing an unseen edge can change every entry of $(I + L_G)^{-1}$. Suppose only q labeled vertices, called *ports*, are visible to a distributed processor. The rest of the graph and its input coordinates are hidden. The hidden network may be arbitrarily large: a modest q means a modest observed interface, not a small physical graph. Our objective is not topology recovery; it is to select one port-only linear map that works for every admissible hidden completion.

The apparent uncertainty set is infinite. Hidden topology, cycles, weights, and input dimension all vary, and arbitrary

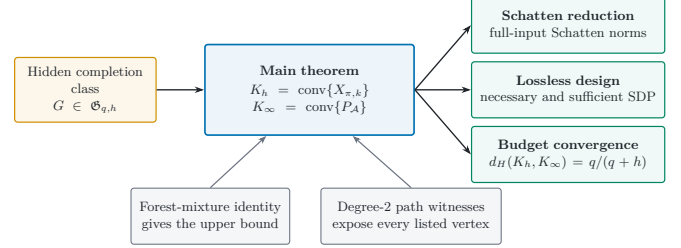


Fig. 1. Logical structure of the result. The forest-mixture identity supplies the upper inclusion, while connected maximum-degree-two paths expose the listed vertices and give the converse. The exact completion hulls then imply the three application corollaries.

large or small conductances rule out a direct perturbation-radius description. The central result replaces this family exactly by finitely many boundary scenarios. For a finite hidden budget they are indexed by a full set partition of the ports and an integer budget allocation. With no budget they are indexed by partial partitions. The resulting algorithm is fixed-parameter in the port count and independent of the hidden-network cardinality, but it is not polynomial in q .

Our contributions are fourfold.

- We characterize the exact finite- h completion hull, all of its exposed vertices, and its unbounded partial-partition limit. Connected weighted paths give the converse, so cycles and high degree do not create missing extremes.
- We retain the complete hidden-input operator and reduce its worst-case unitarily invariant norm to canonical finite scenarios. The unbounded Schatten reduction holds exactly if and only if the exponent is at least two.
- We convert spectral robust design under zero patterns, r -hop support, shared coefficients, or any semidefinite-representable local rule into a finite necessary-and-sufficient SDP, with exact removal of nonvertex scenarios.
- We validate both the certificate and its computation on a fixed theorem-matched design grid, a same-distribution sampling-budget study, and explicit connected path witnesses. Fixed-known-graph approximation and hardware timing are kept as secondary stress tests.

The proof uses the classical matrix-forest representation and the established determinantal mean-projection identity [1]–[4]. We claim neither identity as new. The new object is their exact port image under a hidden-node budget and its consequences for common local-filter design.

II. RELATED WORK AND DISTINCTION

Graph filters express linear signal processing through functions of a graph shift or Laplacian [5]–[7]. Distributed implementations commonly approximate those functions by polynomial or rational recursions on a *known* communication graph [8], [9]. Off-diagonal decay of sparse matrix functions also motivates local inverse approximations [10], [11]. Our setting is different: the target resolvent itself varies with an unrestricted hidden completion, while the same port operator must be used throughout.

Robust graph signal processing has treated random graph shifts, topology perturbations, and robust graph-filter identification [12]–[14]. Hidden-variable graph learning instead infers latent interactions from observed signals [15]. These approaches impose probabilistic, perturbative, or identification structure. Here no nominal hidden graph is available and no completion is estimated. Likewise, generic scenario optimization gives sample-dependent probabilistic guarantees [16]; our partition scenarios are deterministic and necessary and sufficient for the stated completion class.

Normalized partition matrices, electrical boundary partitions, and Kron elimination provide closely related geometry [17]–[20]. To our knowledge, those results do not give the joint finite-hidden-budget hull, exposed-vertex rule, connected path converse, full-hidden-input reduction, and exact local-rule SDP proved below. This distinction is an “as far as we are aware” statement, not a claim of priority for the forest identity or the neighboring partition constructions.

III. COMPLETION MODEL AND LOCAL RULES

Let $\Gamma = \{1, \dots, q\}$ be the labeled port set. A finite completion is a weighted undirected graph

$$G = (V, E, w), \quad V = \Gamma \dot{\cup} Z, \quad w_e \geq 0,$$

where Z is a set of hidden vertices and L_G is the weighted graph Laplacian. All vertices have unit grounding. Define

$$\begin{aligned} T_G &= (I_{q+|Z|} + L_G)^{-1}, \\ E_\Gamma &= [I_q \ 0], \quad X_G = E_\Gamma T_G E_\Gamma^T. \end{aligned} \quad (2)$$

For $h \in \mathbb{Z}_{\geq 0}$, let $\mathfrak{G}_{q,h}$ contain every such completion with $|Z| \leq h$, with no restriction on cycles or finite edge conductances. Let $\mathfrak{G}_{q,\infty} = \bigcup_{h \geq 0} \mathfrak{G}_{q,h}$. Write $\mathfrak{G}_{q,h}^\pm$ for the subclass with $|Z| = h$, and $\mathfrak{G}_{q,h}^\pm$ for its connected members.

All closures below are in the finite-dimensional space $\mathbb{S}^q$ with the operator-norm topology. Thus an equality involving a closure describes limiting high-conductance or weak-link graphs; it does not assert that every boundary point is attained at finite conductance. Likewise, $\sup_G$ always means a supremum over the stated completion class, rather than a maximum on a fixed topology.

Fix an output matrix $C \in \mathbb{R}^{d \times q}$. The centralized port output on a full input $y \in \mathbb{R}^{q+|Z|}$ is $CE_\Gamma T_G y$. A common local linear rule can read only $E_\Gamma y$ and has the form

$$QE_\Gamma y, \quad Q \in \mathcal{Q}_{\text{loc}} \subseteq \mathbb{R}^{d \times q}. \quad (3)$$

The full-input error operator is

$$A_G(Q) = CE_\Gamma T_G - QE_\Gamma \in \mathbb{R}^{d \times (q+|Z|)}. \quad (4)$$

The word “common” is essential: Q cannot depend on the hidden completion. In graph-filter language, T_G is the global rational filter, C selects or combines port outputs, and Q is its completion-independent local surrogate. The spectral norm of (4) is the worst-case signal-energy gain from the complete input, including signals injected at hidden vertices.

A convenient general representation of local implementability is

$$Q(\theta) = Q_0 + \sum_{a=1}^s \theta_a B_a, \quad \theta \in \Theta, \quad (5)$$

where Θ is closed and semidefinite representable. Fixed zero patterns are obtained by omitting coordinate matrices B_a . If output row i is located at port $o(i)$ in a visible communication graph H , an r -round rule imposes

$$Q_{ij} = 0 \quad \text{when} \quad \text{dist}_H(o(i), j) > r.$$

Here H is a fixed protocol overlay on the labeled ports, specified before the unknown physical completion G and independent of it. Equating coefficients of selected B_a ’s gives shared-coefficient or shared-opcode rules. These restrictions may be intersected with unbiasedness, row-sum, or other affine constraints. No particular choice of $\mathcal{Q}_{\text{loc}}$ is needed for the geometric result.

A. Finite and unbounded scenario sets

For a full set partition π of Γ and an allocation $k = (k_B)_{B \in \pi} \in \mathbb{Z}_{\geq 0}^{|\pi|}$ with $|k|_1 = \sum_B k_B \leq h$, set

$$\begin{aligned} X_{\pi,k} &= \sum_{B \in \pi} \frac{\mathbf{1}_B \mathbf{1}_B^T}{|B| + k_B}, \\ \mathcal{X}_{q,h} &= \{X_{\pi,k} : \pi \vdash \Gamma, |k|_1 \leq h\}, \\ K_h &= \text{conv } \mathcal{X}_{q,h}. \end{aligned} \quad (6)$$

Here $\mathbf{1}_B$ is the indicator of B in $\mathbb{R}^q$. Every $X \in \mathcal{X}_{q,h}$ is a positive contraction, but it is generally not a projection. We write $\pi \vdash U$ to mean that π is a set partition of U .

For a subset $U \subseteq \Gamma$ and a partition π of U , define

$$\begin{aligned} P_B &= \frac{\mathbf{1}_B \mathbf{1}_B^T}{|B|}, \quad P_{\pi,U} = \sum_{B \in \pi} P_B, \\ \mathcal{P}_q^\partial &= \{P_{\pi,U} : U \subseteq \Gamma, \pi \vdash U\}, \\ K_\infty &= \text{conv } \mathcal{P}_q^\partial. \end{aligned} \quad (7)$$

The empty active set gives $P = 0$. Every member of $\mathcal{P}_q^\partial$ is an orthogonal projection. Joining all inactive ports to one cemetery symbol gives a bijection with partitions of $q+1$ elements, so $|\mathcal{P}_q^\partial| = B_{q+1}$, where B_j is a Bell number.

IV. EXACT COMPLETION GEOMETRY

We now state the single geometric theorem used throughout the paper. It includes the finite physical model and its unbounded sharp limit.

Theorem 1 (Finite-budget completion hull and partial-partition limit). *For every $q \geq 1$ and $h \geq 0$, the following statements hold.*

- (i) *The finite-budget port image has the exact closed convex hull*

$$\text{cl conv}\{X_G : G \in \mathfrak{G}_{q,h}\} = K_h. \quad (8)$$

The same right-hand side is obtained for either exact-budget class:

$$\begin{aligned} \text{cl conv}\{X_G : G \in \mathfrak{G}_{q,h}^{\bar{c}}\} &= K_h, \\ \text{cl conv}\{X_G : G \in \mathfrak{G}_{q,h}^{\bar{c},c}\} &= K_h. \end{aligned} \quad (9)$$

Every $X_{\pi,k} \in \mathcal{X}_{q,h}$ is approached by a sequence in $\mathfrak{G}_{q,h}^{\bar{c},c}$ consisting of weighted paths, hence of graphs with maximum degree at most two.

- (ii) *The parameterization in (6) has no repetitions, and its cardinality is*

$$N(q, h) = \sum_{s=1}^q S(q, s) \binom{h+s}{s}, \quad (10)$$

where $S(q, s)$ is a Stirling number of the second kind. If $h \geq 1$, the vertices of K_h are exactly

$$\text{vert}(K_h) = \{X_{\pi,0} : \pi \vdash \Gamma\} \cup \{X_{\pi,k} : |k|_1 = h\}, \quad (11)$$

and every listed vertex is exposed. Consequently,

$$N_{\text{vert}}(q, h) = \sum_{s=1}^q S(q, s) \left[1 + \binom{h+s-1}{s-1} \right]. \quad (12)$$

For $h = 0$, all B_q full-partition projections are exposed vertices.

- (iii) *The hulls satisfy $K_h \subseteq K_\infty$ and*

$$\text{cl conv}\{X_G : G \in \mathfrak{G}_{q,\infty}\} = K_\infty. \quad (13)$$

Every $P \in \mathcal{P}_q^\partial$ is an exposed vertex of K_∞ and is approached by connected weighted paths.

- (iv) *With operator norm as the metric, the convergence is exact for every q and h :*

$$d_H^{\text{op}}(K_h, K_\infty) = \frac{q}{q+h}. \quad (14)$$

Proof. Forest-mixture upper bound. Give G an arbitrary orientation and write $L_G = AA^T$, where the column for edge $e = (u, v)$ is $\sqrt{w_e}(e_u - e_v)$. The established L -ensemble mean-projection identity gives

$$(I + AA^T)^{-1} = \mathbb{E} P_{\ker A_S^T}, \quad \Pr(S) = \frac{\det(A_S^T A_S)}{\det(I + A^T A)}, \quad (15)$$

with the empty determinant equal to one [3], [4]. A dependent edge set has zero probability; for an incidence matrix, the remaining sets are precisely forests. If the components of such a forest are D , then

$$P_{\ker A_S^T} = \sum_D \frac{\mathbf{1}_D \mathbf{1}_D^T}{|D|}. \quad (16)$$

For every component meeting Γ , put $B = D \cap \Gamma$ and $k_B = |D \cap Z|$. These nonempty B 's form a full partition of Γ , $\sum_B k_B \leq |Z| \leq h$, and the port principal block of (16) is exactly $X_{\pi,k}$. Hidden-only components consume budget but contribute no port entry. Taking the port block in (15) proves $X_G \in K_h$ and

hence the upper inclusion in (8). This is the only place where the known forest-mixture identity is used.

Connected path converse. Fix (π, k) . For each $B \in \pi$, arrange its ports and k_B hidden vertices into one path segment. If exactly h hidden vertices are required, arrange the unused vertices in additional hidden-only segments. Give every within-segment edge conductance t . At zero conductance between segments,

$$(I + tL_F)^{-1} \longrightarrow P_{\ker L_F} \quad (t \rightarrow \infty), \quad (17)$$

whose port block is $X_{\pi,k}$. Concatenate all segments at their endpoints with positive bridge conductances ε . The resulting finite graph is a connected path. For each sufficiently large t_n , continuity of the inverse permits a sufficiently small $\varepsilon_n > 0$ so that the same limit is obtained. This diagonal sequence proves the reverse inclusion, all three completion-class variants, and the degree-two assertion. Closure is in general necessary: $T_G \succ 0$ at finite conductance, whereas many scenario matrices are singular.

Counting and absence of repetitions. For a partition with s blocks, the number of weak allocations with total at most h is $\binom{h+s}{s}$. Summing over the $S(q, s)$ partitions proves (10). The positive off-diagonal pattern of a scenario recovers its blocks. On a recovered block B , the common value $1/(|B| + k_B)$ recovers k_B , so no two parameters give the same matrix.

Finite exposed vertices. Suppose $0 < |k|_1 < h$ and choose B with $k_B > 0$. Holding all other allocations fixed, set

$$K = h - \sum_{D \neq B} k_D > k_B.$$

Because $1/(|B| + k_B)$ lies strictly between $1/|B|$ and $1/(|B| + K)$, the scenario is a strict convex combination of the two feasible scenarios obtained by replacing k_B with 0 and K . It is not a vertex.

For $k = 0$, $X_{\pi,0} = P_\pi$ is an orthogonal projection. The functional $X \mapsto \langle 2P_\pi - I, X \rangle$ uniquely maximizes at P_π over the entire positive-contraction interval $0 \preceq X \preceq I$: indeed,

$$\begin{aligned} \langle 2P_\pi - I, X \rangle &= \text{tr}(P_\pi X P_\pi) \\ &\quad - \text{tr}((I - P_\pi)X(I - P_\pi)) \\ &\leq \text{rank } P_\pi. \end{aligned} \quad (18)$$

and equality forces the two diagonal compressions to be P_π and zero; positivity of both X and $I - X$ then forces the cross terms to vanish. Thus $X = P_\pi$, which exposes it in K_h .

It remains to expose a target (π, k^*) with $|k^*|_1 = h$. Define a symmetric separator W_π by

$$\begin{aligned} (W_\pi)_{ii} &= -(|B| - 1), & i \in B, \\ (W_\pi)_{ij} &= \begin{cases} 1, & i \neq j, i, j \in B, \\ -1, & i \in B, j \in D, B \neq D, \end{cases} & B, D \in \pi. \end{aligned}$$

If $S \neq \emptyset$ and $r_B = |S \cap B|$, then

$$\begin{aligned} \mathbf{1}_S^T W_\pi \mathbf{1}_S &= \sum_{B \in \pi} r_B(r_B - |B|) \\ &\quad - 2 \sum_{B < D} r_B r_D \leq 0, \end{aligned} \quad (19)$$

with equality exactly when S is one target block. Hence $\langle W_\pi, X_{\pi,k} \rangle = 0$ for the target partition and is strictly negative for every other partition.

On each target block set

$$A_B = -\frac{\gamma_B}{|B|^2} \mathbf{1}_B \mathbf{1}_B^T, \quad \gamma_B = (|B| + k_B^*)^2,$$

and put all cross-block entries of A to zero. Thus $\mathbf{1}_B^T A_B \mathbf{1}_B = -\gamma_B$. For the target partition,

$$f(k) = \langle A, X_{\pi,k} \rangle = -\sum_B \frac{\gamma_B}{|B| + k_B}. \quad (20)$$

The gain from the $(r+1)$ st hidden vertex in block B is

$$\Delta_B(r) = \frac{\gamma_B}{(|B| + r)(|B| + r + 1)}.$$

Selected increments have gain greater than one and unselected increments have gain less than one. Hence k^* uniquely maximizes (20) under $|k|_1 \leq h$: any different feasible allocation omits a selected increment or replaces one by a strictly smaller unselected increment. If π is the only full partition (which occurs only for $q = 1$), this already shows that A exposes the target. Otherwise let

$$\delta = -\max_{\substack{\tilde{\pi} \neq \pi, \tilde{k} \\ |k|_1 \leq h}} \langle W_\pi, X_{\tilde{\pi}, \tilde{k}} \rangle > 0, \quad R = \max_{X \in \mathcal{X}_{q,h}} |\langle A, X \rangle|.$$

For $M > 2R/\delta$, $MW_\pi + A$ ranks every wrong partition below every target-partition candidate, and A uniquely selects k^* within the latter. Thus $MW_\pi + A$ exposes the target. This proves (11); stars-and-bars gives (12).

Partial-partition limit and exposedness. Write each finite scenario as

$$X_{\pi,k} = \sum_{B \in \pi} \alpha_B P_B, \quad \alpha_B = \frac{|B|}{|B| + k_B} \in (0, 1].$$

Independently retain block B with probability α_B . The resulting random sum is a partial-partition projection with expectation $X_{\pi,k}$. Therefore $K_h \subseteq K_\infty$.

Conversely, fix $P_{\pi,U}$ and let $I = \Gamma \setminus U$. If $I \neq \emptyset$, append I as one full-partition block and allocate all h hidden vertices to it. This gives

$$X_h = P_{\pi,U} + \frac{|I|}{|I| + h} P_I \longrightarrow P_{\pi,U}. \quad (21)$$

If I is empty, take $X_h = P_{\pi,U}$. Applying the connected path construction above and taking a diagonal sequence proves (13) and its path converse. For distinct orthogonal projections $P, P' \in \mathcal{P}_q^\partial$,

$$\begin{aligned} \langle 2P - I, P' \rangle &= \text{tr } P - \|P - P'\|_F^2 \\ &< \text{tr } P = \langle 2P - I, P \rangle, \end{aligned} \quad (22)$$

so every partial scenario is exposed.

Exact Hausdorff distance.

Equation (21) gives $\|X_h - P_{\pi,U}\|_2 \leq q/(q+h)$. Extending this approximation from vertices to their convex combinations

proves the upper bound in (14). For the reverse bound, let $u = \mathbf{1}_\Gamma / \sqrt{q}$. Every finite scenario satisfies

$$u^T X_{\pi,k} u = \frac{1}{q} \sum_{B \in \pi} \frac{|B|^2}{|B| + k_B} \geq \frac{1}{q} \frac{(\sum_B |B|)^2}{\sum_B (|B| + k_B)} \geq \frac{q}{q+h}. \quad (23)$$

The inequality persists under convex combinations, while the one-block scenario with $k = h$ has operator norm exactly $q/(q+h)$. Since $0 \in K_\infty$, its distance to K_h is exactly this value. The proof is complete. $\square$

V. FULL HIDDEN-INPUT ERROR AND THE SCHATTEN BOUNDARY

For $X \in \mathcal{X}_{q,h}$ define the canonical error

$$\mathcal{E}_Q(X) = [CX - Q, C(X - X^2)^{1/2}] \in \mathbb{R}^{d \times 2q}. \quad (24)$$

The square root exists because $0 \preceq X \preceq I$. Its second block is not optional: it records exactly the hidden-input energy absent from the port principal block. For later use, define the affine map and its Gram completion

$$\Psi_Q(X) := CX - Q - CXQ^T - QXC^T, \quad (25)$$

$$M_Q(X) := \Psi_Q(X) + QQ^T = \mathcal{E}_Q(X) \mathcal{E}_Q(X)^T. \quad (26)$$

A unitarily invariant norm below means a zero-padding-consistent family over rectangular sizes, equivalently one fixed symmetric gauge applied to every finite singular-value sequence. This convention is needed because the input width varies with the hidden completion.

Corollary 2 (Exact full-input Schatten reductions). *Fix $C \in \mathbb{R}^{d \times q}$ and $Q \in \mathbb{R}^{d \times q}$.*

(i) *For every finite h and every $1 \leq s \leq \infty$,*

$$\sup_{G \in \mathfrak{G}_{q,h}} \|\mathcal{A}_G(Q)\|_{S_s} = \max_{X \in \mathcal{X}_{q,h}} \|\mathcal{E}_Q(X)\|_{S_s}. \quad (27)$$

The same equality holds for every unitarily invariant matrix norm.

(ii) *If $2 \leq s \leq \infty$, the finite maximum in (27) may be restricted to the vertices in (11). In the unbounded model,*

$$\sup_{G \in \mathfrak{G}_{q,\infty}} \|\mathcal{A}_G(Q)\|_{S_s} = \max_{P \in \mathcal{P}_q^\partial} \|CP - Q\|_{S_s}. \quad (28)$$

(iii) *The exponent threshold in (28) is sharp over the Schatten family: a universal equality for all C, Q and all completions holds if and only if $s \geq 2$.*

Proof. For the forest random projection $\hat{P} = P_{\ker A_S^T}$ in (15), put $A_{\hat{P}} = CE_\Gamma \hat{P} - QE_\Gamma$. Convexity of any matrix norm gives

$$\|\mathcal{A}_G(Q)\| = \|\mathbb{E} A_{\hat{P}}\| \leq \mathbb{E} \|A_{\hat{P}}\|.$$

With $X = E_\Gamma \hat{P} E_\Gamma^T$ and $\hat{P}^2 = \hat{P}$, direct multiplication gives the key affine Gram identity

$$A_{\hat{P}} A_{\hat{P}}^T = M_Q(X) = \mathcal{E}_Q(X) \mathcal{E}_Q(X)^T. \quad (29)$$

Thus $A_{\hat{P}}$ and $\mathcal{E}_Q(X)$ have the same singular values. Every forest compression is in $\mathcal{X}_{q,h}$, proving the upper bound in (27). For a fixed finite candidate, use the exact-budget connected

path construction in Theorem 1. If $R_n = E_\Gamma T_{G_n}$ is its port-row operator, the construction can be diagonalized so that $E_\Gamma T_{G_n}^2 E_\Gamma^T \rightarrow X$ and $X_{G_n} \rightarrow X$. Consequently,

$$\mathcal{A}_{G_n}(Q)\mathcal{A}_{G_n}(Q)^T \rightarrow M_Q(X), \quad (30)$$

which proves the reverse inequality for every zero-padding-consistent unitarily invariant norm without identifying matrices of changing width.

For finite $2 \leq s < \infty$, the s th power of the scenario loss is

$$X \mapsto \text{tr} \left(M_Q(X)^{s/2} \right).$$

The scalar trace functional $Z \mapsto \text{tr}(Z^{s/2})$ is convex on the positive-semidefinite cone [21], and M_Q is affine in X . For $s = \infty$, use the convex function $\lambda_{\max}(M_Q(X))$. A convex function on K_h has a maximizer among its vertices, proving the pruning statement. Every finite scenario belongs to $K_\infty = \text{conv } \mathcal{P}_q^\partial$; affinity of M_Q and the same convexity bound its loss by the largest partial-scenario loss. Conversely, every partial scenario is approached by a changing-budget Gram sequence. For $P = P_{\pi,U}$, use X_h from (21), realize it by an exact- h path, and choose the conductances so that its physical error Gram is within $1/h$ of $M_Q(X_h)$. Since $M_Q(X_h) \rightarrow M_Q(P)$ and $P^2 = P$, the limiting singular values are those of $[CP - Q \ 0]$. This proves (28) without zero-padding matrices whose column dimensions differ.

To prove sharpness, take one port, one hidden vertex, and let their connecting conductance tend to infinity. The full projection tends to

$$\hat{P} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Choose $C = (1, 1)^T$ and $Q = (1, 0)^T$. The two partial scenarios $P = 0, 1$ have error vectors $-Q$ and $C - Q$, both of Schatten norm one. The full limiting error is

$$\begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix},$$

whose two singular values are $1/\sqrt{2}$. Its Schatten- s norm is $2^{1/s-1/2} > 1$ exactly when $1 \leq s < 2$. The positive result includes $s = 2$, so the boundary is exact. $\square$

For $s < 2$, equation (27) remains valid, but an unexposed finite candidate cannot be deleted merely because it is not a vertex of K_h . Explicitly, let $q = 1$, $h = 2$, $C = (1, 1)^T$, and $Q = (1, 0)^T$. The candidates are $x \in \{1, 1/2, 1/3\}$, with finite-hull vertices 1 and $1/3$, and

$$M_Q(x) = \text{diag}(1 - x, x).$$

The Schatten- s loss to the s th power is $(1 - x)^{s/2} + x^{s/2}$. For every $1 \leq s < 2$ it is strictly maximized at the nonvertex $x = 1/2$.

VI. LOSSLESS LOCAL-RULE SDP

Let $\mathcal{V}_{q,h}$ denote the finite vertex set in (11), with all full partitions used when $h = 0$. The following certificate directly supports zero-pattern, r -hop, and shared coefficient constraints through (5).

Corollary 3 (Necessary-and-sufficient spectral SDP). *Assume $\mathcal{Q}_{\text{loc}}$ is nonempty, closed, and semidefinite representable. For the finite completion class, the squared optimal radius is exactly the optimal value of*

$$\begin{aligned} & \underset{Q \in \mathcal{Q}_{\text{loc}}, \rho \geq 0}{\text{minimize}} && \rho \\ & \text{subject to} && \begin{bmatrix} \rho I_d - \Psi_Q(X) & Q \\ Q^T & I_q \end{bmatrix} \succeq 0, \quad X \in \mathcal{V}_{q,h}. \end{aligned} \quad (31)$$

Equivalently, for any prescribed $\varepsilon \geq 0$, a common local Q with $\sup_{G \in \mathfrak{G}_{q,h}} \|\mathcal{A}_G(Q)\|_2 \leq \varepsilon$ exists if and only if (31) is feasible with $\rho = \varepsilon^2$.

For the unbounded class, the optimal radius is exactly the optimal value of

$$\begin{aligned} & \underset{Q \in \mathcal{Q}_{\text{loc}}, \tau \geq 0}{\text{minimize}} && \tau \\ & \text{subject to} && \begin{bmatrix} \tau I_d & CP - Q \\ (CP - Q)^T & \tau I_q \end{bmatrix} \succeq 0, \quad P \in \mathcal{P}_q^\partial. \end{aligned} \quad (32)$$

Both optima are attained. There is no completion-scenario relaxation gap.

Proof. For fixed X , the Schur complement of the lower-right identity in (31) is

$$\rho I_d - M_Q(X) \succeq 0.$$

By (29), this is equivalent to $\|\mathcal{E}_Q(X)\|_2^2 \leq \rho$. Corollary 2 and exact vertex pruning prove necessity and sufficiency. For a partial projection, the hidden block in (24) vanishes; the standard rectangular spectral-norm epigraph gives (32) [22]. The scenario families are finite. Moreover $X = I$ is a finite scenario and $P = 0$ is an unbounded scenario, so bounded objective values bound Q ; closedness gives attainment. $\square$

The finite formulation uses $N_{\text{vert}}(q, h)$ LMIs of order $d + q$. The unbounded formulation uses B_{q+1} LMIs of the same order. This is an exact boundary-size fixed-parameter algorithm, not a polynomial-time claim in q . The accompanying implementation enumerates restricted-growth set partitions, weak integer compositions, and active scenarios, and it verifies the returned Q against every exact candidate after solving.

For fixed q , the unbounded SDP covers every finite hidden-network size; its online rule stores neither the hidden topology nor a full-graph inverse.

VII. QUANTITATIVE BUDGET HIERARCHY

Define the fixed-rule spectral radii

$$\begin{aligned} R_h(Q) &= \sup_{G \in \mathfrak{G}_{q,h}} \|\mathcal{A}_G(Q)\|_2, \\ R_\infty(Q) &= \sup_{G \in \mathfrak{G}_{q,\infty}} \|\mathcal{A}_G(Q)\|_2. \end{aligned}$$

and let $L_Q = \|C\|_2^2 + 2\|C\|_2\|Q\|_2$.

Corollary 4 (Finite-budget convergence of certified risk). *For every fixed C, Q and every $h \geq 0$,*

$$0 \leq R_\infty(Q)^2 - R_h(Q)^2 \leq L_Q \frac{q}{q+h}. \quad (33)$$

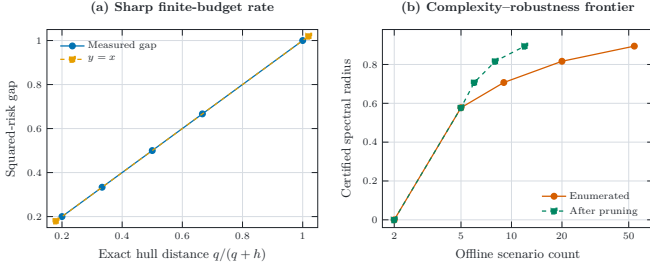


Fig. 2. Finite hidden-budget hierarchy for the two-port fixed-rule audit $C = Q = 2^{-1/2}[1 \ 1]$. The measured finite-to-unbounded squared-risk gaps coincide numerically with the exact distance $q/(q+h)$. Increasing the hidden budget increases both the certified radius and the offline scenario count; exact pruning deletes redundant LMIs without changing the radius.

Proof. Let $g_Q(X) = \lambda_{\max}(M_Q(X))$, with M_Q as in (29). Since $K_h \subseteq K_\infty$ and g_Q is convex, the left inequality follows from the exact scenario reductions. For $X, Y \in K_\infty$, Weyl's inequality and submultiplicativity give

$$|g_Q(X) - g_Q(Y)| \leq \|M_Q(X) - M_Q(Y)\|_2 \leq L_Q \|X - Y\|_2.$$

Combine this Lipschitz bound with the exact Hausdorff distance (14). $\square$

VIII. GRAPH-SMOOTHING EXAMPLES AND ALGORITHMIC MEANING

A. A finite-budget gain that is not imposed by a protocol constraint

Let $q = 2$, $h = 1$, and $C = P_{\text{av}} = \mathbf{1}\mathbf{1}^T/2$. The five exact candidates are

$$I, \quad \text{diag}(1/2, 1), \quad \text{diag}(1, 1/2), \\ P_{\text{av}}, \quad \frac{2}{3}P_{\text{av}}.$$

The unconstrained solution of (31) is

$$Q_* = \frac{2}{3}P_{\text{av}}, \quad R_1^* = \frac{\sqrt{2}}{3} = 0.471404 \dots \quad (34)$$

At this Q , the scenario radii are

$$\frac{1}{3}, \quad \frac{\sqrt{7}}{6}, \quad \frac{\sqrt{7}}{6}, \quad \frac{1}{3}, \quad \frac{\sqrt{2}}{3}.$$

The last scenario has the unavoidable hidden-input term

$$C \left(\frac{2}{3}P_{\text{av}} - \frac{4}{9}P_{\text{av}} \right) C^T = \frac{2}{9}P_{\text{av}},$$

which proves optimality. The unbounded optimum is $1/2$, so knowing $h = 1$ reduces radius by 5.72% and squared risk by 11.11%. At $h = 0$, the choice $Q = C$ has zero error.

B. Why unbounded design needs locality

If Q is unrestricted, $P = 0$ and $P = I$ imply

$$\max_{P \in \mathcal{P}_q^0} \|CP - Q\|_2 \geq \frac{1}{2} \|C\|_2,$$

while $Q = C/2$ attains equality for every orthogonal projection P . Thus the unbounded free- Q problem has the transparent solution $Q_* = C/2$, $R_\infty^* = \|C\|_2/2$. Communication sparsity

or another real protocol restriction is required for a substantive design tradeoff.

For a minimal shared-opcode example, take

$$C = \text{diag}(1, 2), \quad \mathcal{Q}_{\text{loc}} = \{\alpha I_2 : \alpha \in \mathbb{R}\}.$$

The exact five-scenario problem gives

$$\alpha_* = \frac{4}{3} - \frac{2\sqrt{10}}{15} = 0.911696 \dots, \\ R_* = \frac{10+2\sqrt{10}}{15} = 1.088304 \dots \quad (35)$$

Dropping the connected partial scenario P_{av} instead gives the incorrect answer $\alpha = 1$, $R = 1$: the robust radius is underestimated by 8.83%. This example also illustrates why the solver reports the active scenario set rather than only the optimal matrix.

C. Exact geometry versus a generic positive-contraction set

The common surrogate $0 \preceq X \preceq I$ is safe because it contains K_∞ , but it need not be tight. This difference is already analytic for $q = 2$, $d = 1$, and the constant-preserving family

$$C = [0 \ 2], \quad Q(a) = [a \ 2-a]. \quad (36)$$

The exact five partial-partition scenarios give

$$R_{\text{exact}}(a) = \max\{[a^2 + (2-a)^2]^{1/2}, \\ \sqrt{2}|a|, \sqrt{2}|1-a|\}. \quad (37)$$

Its optimum is $\sqrt{2}$ at $a = 1$. Maximizing the same affine Gram loss over the full positive-contraction interval instead gives

$$R_{[0,1]}(a) = 1 + \sqrt{a^2 + (1-a)^2}, \quad (38)$$

whose optimum is $1 + 1/\sqrt{2}$ at $a = 1/2$. Exact terminal geometry therefore lowers the optimal certified radius by 17.2% in this example. At the network-optimal rule, the generic certificate is 2 rather than $\sqrt{2}$, a 41.4% overestimate. This example isolates geometric conservatism from sampling failure and solver error.

D. Offline and online costs

Scenario generation, conic model construction, and SDP solution are offline design costs. They depend exponentially on the port boundary but not on the eventual hidden network size. Once Q is fixed, online estimation is simply $QE_T y$. Under an r -hop zero pattern its operational accounting can therefore be kept separate: r synchronous rounds, protocol-dependent message and byte-hop counts, $\text{nnz}(Q)$ multiply-adds per signal, and storage proportional to $\text{nnz}(Q)$. These quantities are not part of the mathematical radius and should be reported separately in experiments.

IX. COMPUTATIONAL VALIDATION

A. Fixed theorem-matched design grid

All main-paper experiments optimize the same full-input norm as the theorem. The benchmark definition enumerates every cell before solving and reports all of them: $q \in \{3, 4, 5\}$, $h \in \{1, 2, 4\}$, and six rule families per q , for 54 exact design problems. The six families are direct state smoothing

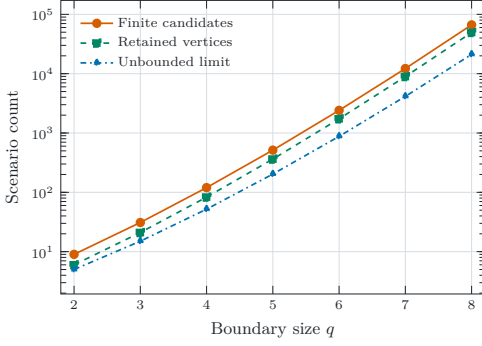


Fig. 3. Exact scenario counts at hidden budget $h = 2$. Finite candidates, retained finite-hull vertices, and unbounded partial partitions all grow combinatorially with q . The reduction is boundary-size fixed-parameter, not polynomial in the boundary size.

($C = I$) with constant preservation $Q\mathbf{1} = \mathbf{1}$ at zero-hop, one-hop, and full-port support on a fixed path overlay H ; heterogeneous diagonal outputs with independent or shared diagonal coefficients; and an unconstrained one-hop state rule as a negative control. For each cell, five $N = 32$ randomized designs use random-cyclic, grid, and geometric completions with log-uniform weights in $[10^{-2}, 10^2]$ and hidden counts cycled through $0, \dots, h$, giving 270 sampled rules. Every sampled rule and every positive-contraction rule is evaluated afterward on the complete exact finite scenario set.

Figure 4 reports the two main comparisons. Across the nine matched (q, h) state-smoothing pairs, one communication round reduces the certified radius over zero-hop operation by a median 18.7%, while full-port support reduces it by 38.8% in the descriptive grid. All nine one-hop solves in the headline comparison have status `optimal`; the zero-hop constraint fixes $Q = I$. The unconstrained negative control is retained in the archive: without constant preservation, the robust center can collapse to a diagonal/no-communication rule, so communication is not beneficial by construction. Over the one-hop constant-preserving subgroup, the generic $0 \preceq X \preceq I$ design has median exact-risk overhead 8.2%; over the full grid it is often tight, which is why the complete distribution rather than only favorable cases is shown.

The representative state problem has $q = 4$, $h = 1$, $C = I$, one-hop path support, and $Q\mathbf{1} = \mathbf{1}$. Its deployed exact rule is

$$Q_* = \begin{bmatrix} .726 & .274 & 0 & 0 \\ .094 & .630 & .276 & 0 \\ 0 & .276 & .630 & .094 \\ 0 & 0 & .274 & .726 \end{bmatrix}, \quad (39)$$

with exact radius 0.808951. The unbounded optimum is 1.000000, so using the known one-node hidden budget improves the radius by 19.1%. The positive-contraction design has exact finite risk 0.891149, or 10.2% overhead after exact reevaluation. Thresholding solver-scale entries at 2×10^{-5} and recertifying gives an implementation with 1 round, 6 scalar-hops (48 byte-hops in double precision), 16 online FLOPs, and 140 bytes of sparse coefficient storage. These are counts for one explicit unicast schedule, not communication lower bounds.

B. Sampling budget and physical path tightness

To avoid a train-test distribution shift, the sampling study draws nested training sets from the same distribution at $N \in \{4, 8, 16, 32, 64, 128, 256\}$. It uses 30 independent seeds for the four-port state problem above and for a three-port, two-hidden-node shared diagonal-opcode problem. Thus there are 420 fitted rules. For a sampled rule Q_N , let $R_N(Q_N)$ be its maximum training radius, $R_h(Q_N)$ its exact finite-budget radius, and R_h^* the robust optimum. We report relative underreporting $[R_h(Q_N) - R_N(Q_N)]/R_h(Q_N)$ and regret $R_h(Q_N)/R_h^* - 1$; a failure means $R_h(Q_N) - R_N(Q_N) > \max\{2 \times 10^{-5}, 10^{-3}R_h(Q_N)\}$. At every sample size and in both cases, all 30 rules fail this test; the per-cell 95% Wilson lower bound on this descriptive failure frequency is 88.6%. At $N = 32$, the state problem has median underreporting 18.0% and median true-risk regret 13.3%. Figure 5 shows that both gaps have an overall downward trend with N but remain nonzero at 256; sampling is therefore a useful design heuristic, not a deterministic certificate.

Finally, we instantiate the converse rather than checking only algebraic scenarios. Three active finite vertices with one, two, and three partition blocks are realized by connected paths containing exactly $h = 1$ hidden node. Within-component conductance is t , bridge conductance is t^{-1} , and t ranges from 10 to 10^6 . At the final value, the largest relative full-input risk gap is 2.09×10^{-6} and the largest port-block operator gap is 2.68×10^{-6} ; every constructed graph is connected and has maximum degree two. Figure 6 makes the physical attainability behind the lower-bound proof numerically visible.

As implementation audits, a separate 135-completion sweep places every sampled physical error below its exact certificate (largest ratio 0.9475), and the pruned and unpruned smoke-test radii agree to 1.7×10^{-9} . At $q = 8$, $h = 2$, generation yields 66,491 candidates, 49,484 retained vertices, and 21,147 partial partitions, exposing the intended boundary-size limit. On the archived platform, all 54 finite designs completed in median 0.168 s and maximum 4.13 s end to end; the slowest cell had $q = 5$, $h = 4$. These one-pass timings are descriptive. Fixed-known-graph local algorithms, PGLib/SuiteSparse scale tests [23], [24], and accelerator timings are reported separately in the supplement because they test a different known-topology task or a particular hardware implementation.

The archived solver accepts C , q , h (or unbounded mode), and affine local rules, and returns Q , its exact-revalidated radius, active scenarios, counts, and timings. The anonymous software supplement contains fixed configurations, records, seeds, provenance, tests, figure sources, and reproduction commands; a permanent repository can replace it after review.

X. DISCUSSION AND SCOPE BOUNDARIES

The following boundaries are part of the result rather than optional qualifications.

- Finite- h equality assumes scalar undirected nonnegative edges and unit grounding. A uniform finite upper bound on conductance can destroy the high-conductance converse, and a positive lower bound on connecting edges can destroy weak-link separation.

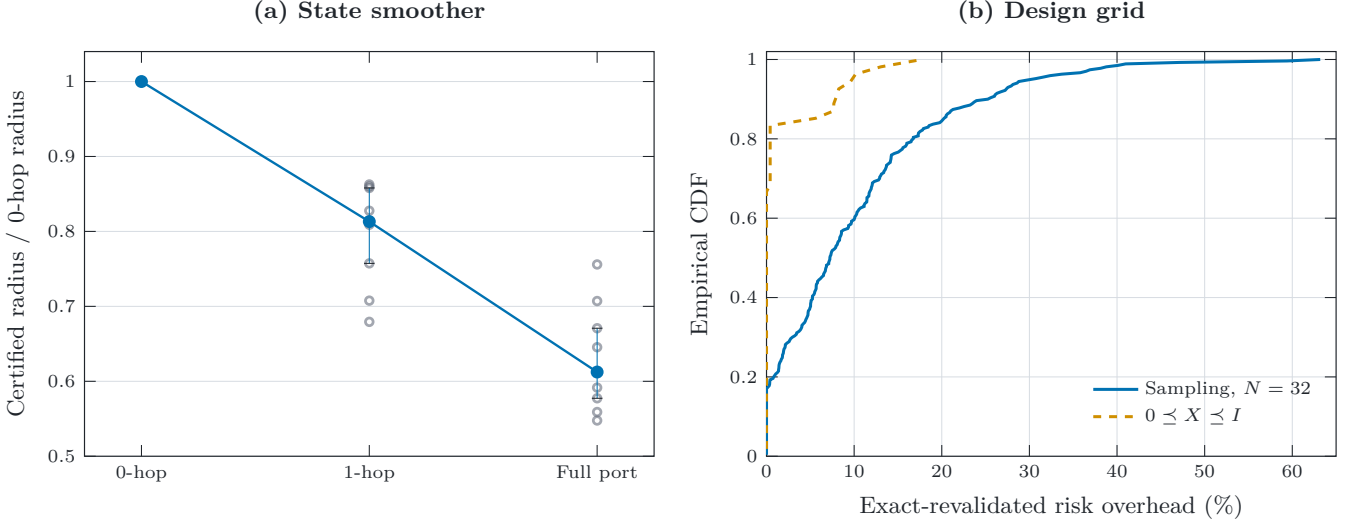


Fig. 4. Fixed theorem-matched design grid. (a) Postsolve-revalidated radii for nine (q, h) pairs under three support classes (27 evaluations), normalized by the corresponding zero-hop radius; gray circles show all pairs and blue markers show median and interquartile range. (b) Descriptive empirical CDFs of exact-revalidated true-risk overhead for 270 $N = 32$ sampled rules and 54 finite-budget evaluations of 18 positive-contraction designs. These grouped grid evaluations are not treated as IID samples.

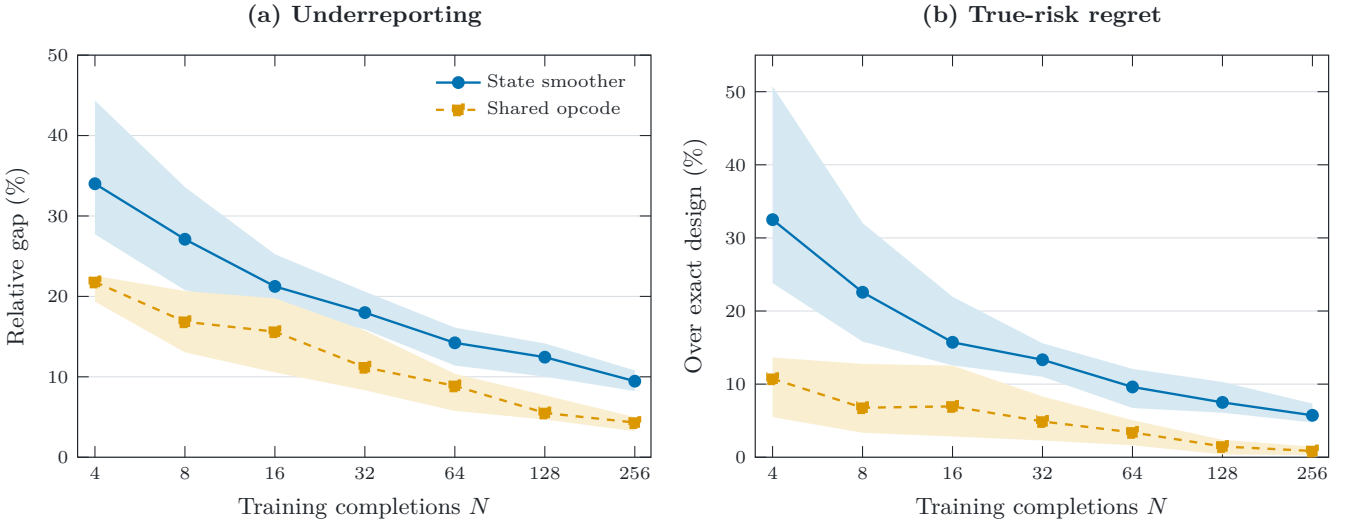


Fig. 5. Same-distribution randomized design with 30 seeds per point; bands are interquartile ranges. (a) Relative gap between exact-revalidated and training radii. (b) Exact-revalidated risk overhead over exact robust design. Training samples are nested within each seed.

- Hidden-input columns cannot be discarded. For finite scenarios they produce $C(X - X^2)C^T$ in (29).
- The protocol overlay H is fixed independently of the physical completion G . If communication edges must instead be inferred from G , the admissible rule class varies with the uncertainty and is a different problem.
- The unbounded partial-partition reduction for full errors is sharp at Schatten exponent two. It is not a theorem for arbitrary convex losses of the full error matrix, and it fails for every Schatten exponent below two.
- The estimator in (1) is graph-regularized state smoothing. A raw WLS measurement-noise transfer includes the measurement operator as an additional right factor; its

MSE is not $\|\mathcal{A}_G(Q)\|_F^2$ without a separate derivation.

- The theorem does not cover directed, signed, block-noncommutative, or nonlinear completion models.

The exact method is most attractive when an arbitrarily large hidden network attaches to a modest port boundary. Bell-number growth in q eventually dominates offline design, as Figure 3 shows; scalable separation, cutting-plane, and symmetry reductions remain outside the present claims.

XI. CONCLUSION

Unknown hidden topology need not be sampled or replaced by a full-block uncertainty set for this graph smoother. Set

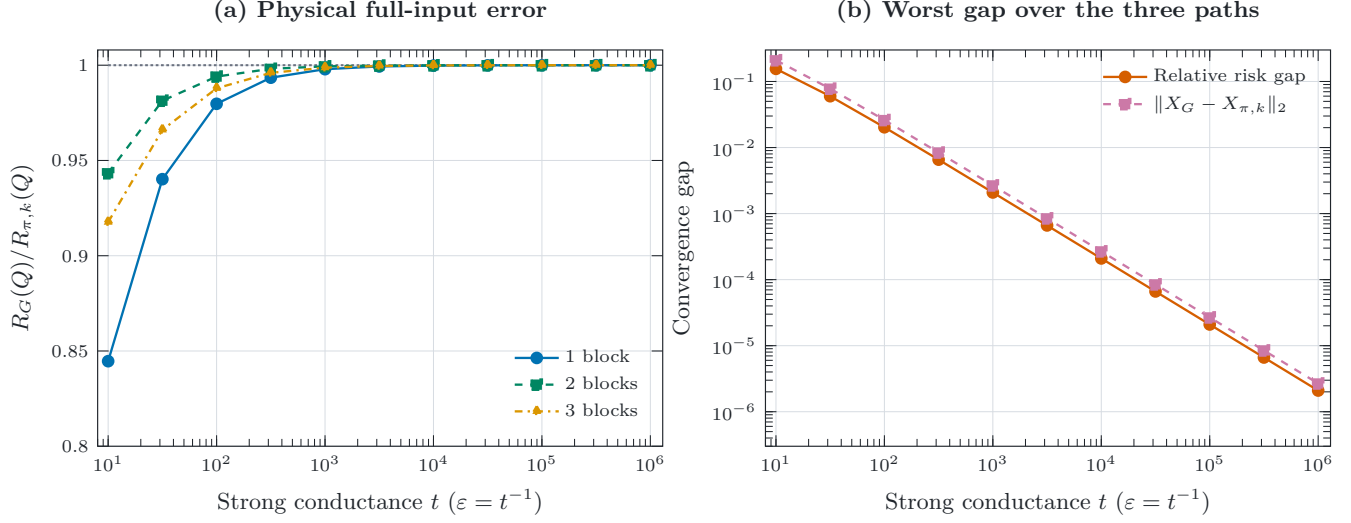


Fig. 6. Connected maximum-degree-two path witnesses for three active exact scenarios. (a) Physical full-input risk approaches its canonical value. (b) Both the worst relative risk gap and port-block operator distance vanish as strong and weak conductances separate.

partitions and hidden-budget allocations give an exact finite description, including the complete hidden input, while connected paths certify that every retained boundary scenario is genuinely sharp. The result supplies a lossless SDP for constrained common local rules and an exact budget-to-unbounded convergence rate. The numerical results separate exact certification from completion sampling and generic positive-contraction relaxation; fixed-graph approximation remains a different task.

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